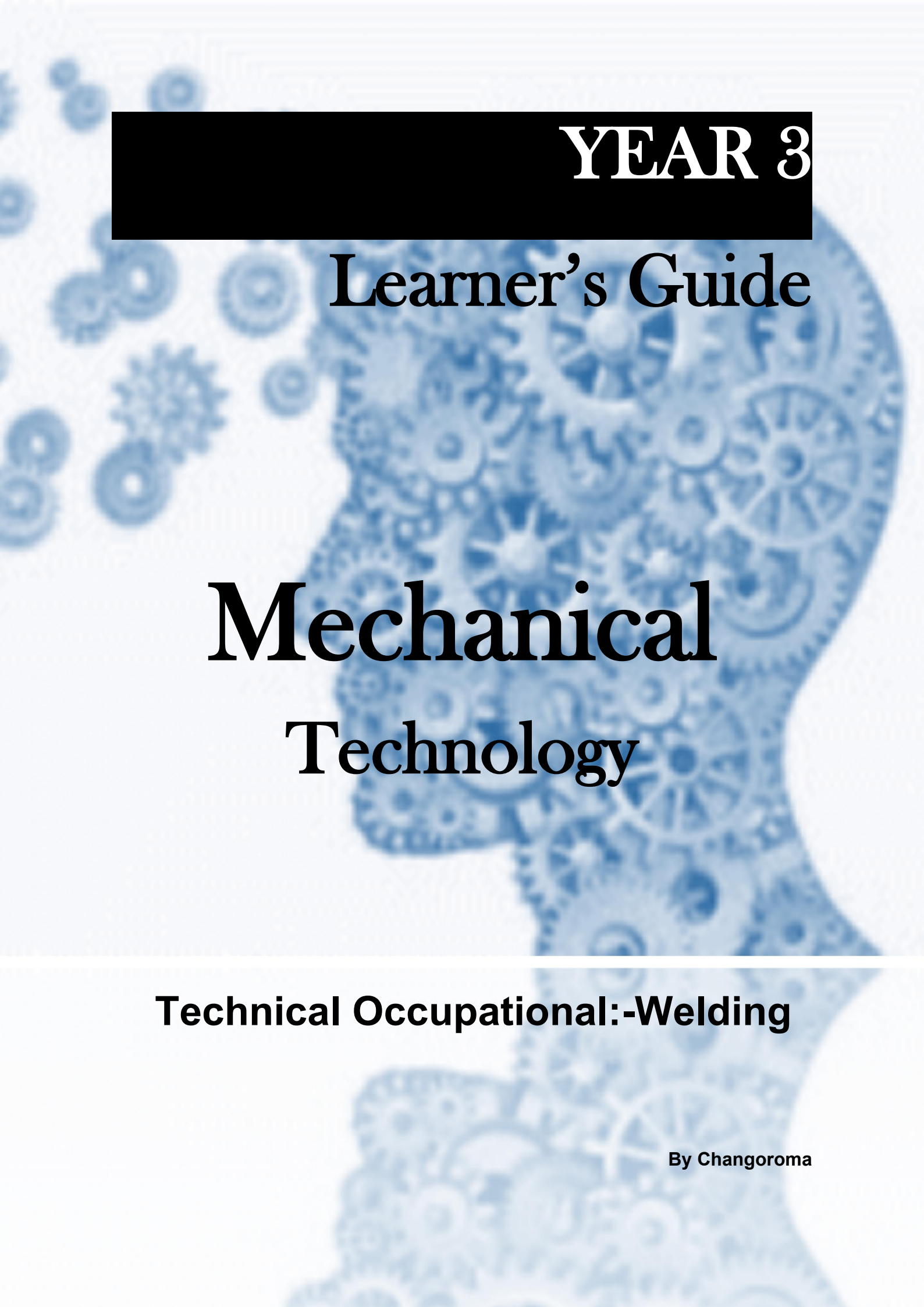




YEAR 3

Learner's Guide

Mechanical Technology

Technical Occupational:-Welding

By Changoroma

SAFETY, HEALTHY & ENVIRONMENT (SHE) RULES

The objective of these instructions are to give education and advice to all employees and learners in safe methods of work and the avoidance of accidents.

It is the policy of a company to ensure that every one works in safe conditions and carries out their tasks in the safe manner. Accidents don't just happen- there's always a cause.

You have a duty to observe the regulations, both in your own interest and that of your fellow workers.

REMEMBER

DOING IT THE CORRECT WAY

IS THE SAFE WAY

The Occupational Health, Safety and Environmental Acts states

(Section 14.a-e, OHSA):-

1. It shall be a duty of every learner in the workshop – to take reasonable care for the health and safety of themselves and other persons who may be affected by their acts or omissions at work.
2. No person shall intentionally or recklessly interfere with or misuse anything provided in the interest of health, safety or welfare in pursuance of any of the relevant statutory provisions (Section 15, OHSA).
3. You are liable to prosecution by the Occupational Health & Safety Inspectorate for failure to observe the above requirements of the Act.
4. Report any item at your workplace that is unsafe, to your area.
5. By doing this you will be helping yourself and making conditions safer for your fellow workers.

The word safety is defined as **FREEDOM FROM DANGER OR RISKS.**

THINK SAFETY and BE SAFE!

REMEMBER, ACCIDENT PREVENTION IS PART OF YOUR JOB

WHY LEARN THE HARD WAY? OBEY SAFETY RULES!

REMEMBER THESE FACTS

- **INCIDENTS** DON'T HAPPEN, THEY **ARE CAUSED**
- THE CAUSE OF 8 OUT 10 INCIDENTS ARE **BY HUMAN**

FAILURE AND NOT MECHANICAL FAILURE

- 9 OUT OF 10 INCIDENTS **ARE PREVENTABLE**
- INCIDENTS DON'T ONLY HAPPEN **TO OTHER PEOPLE**

Occupational Safety

Check yourself! Are you alert, wide-awake and in positive frame of mind to operate a machine?

A machine cannot think, you must do the thinking for it.



Are you taking medication? Does it make you drowsy? Don't operate machinery if you are on medication that can affect your reflexes and reactions to situations.

Keep your mind on the job. Avoid visiting or having unnecessary conversation while operating equipment.



Wear your personal protective equipment whenever required

Eyes		Ears	
Face		Mouth & Nose	
Feet		Hands	
Head		Safety Harness	
Apron		Welding Helmet	

1. Don't wear loose clothing. Ties, rings and watches should not be worn – especially around machinery.
2. Always wear a hard hat in an area where there is a possibility of falling objects.
3. The correct footwear must be worn at all times where areas have been identified on site.
4. When welding always wear welding helmet, leather gloves and leather apron.
5. Always wear safety goggles when grinding and a hood when buffing.

ALWAYS WEAR THE CORRECT PERSONAL PROTECTIVE EQUIPMENT FOR THE JOB.

**Don't come to work
intoxicated or drugged.**



**No food stuffs are permitted in
the workshop**



GENERAL WORKSHOP RULES

- 1) Horseplay- such as throwing things, scuffling, and pointing an air hose at a fellow worker, is dangerous and strictly forbidden.
- 2) Smoking is not permitted anywhere on the premises / school yard.
- 3) Keep your workshop tidy, and your work area clean. Put all waste, rubbish, papers etc. in containers provided.
- 4) Never run in the workshop.
- 5) Cell phones are prohibited in workshop / workstations especially when work is in progress. Switch your cell phone off.
- 6) **Never obstruct walkways with material.**
- 7) **Never try to pull something out of moving rollers, belting or machinery.**
- 8) **Do not start any machine or equipment that has been danger tagged.**
- 9) Before working, with any machine **be sure you understand how it operates.** If you are unsure ask for help.
- 10) If machine is made only for one operator never allow anyone else to help you.
- 11) **Keep hands and fingers clear from danger areas.**
- 12) **Never grind on the side face of a grinding wheel.**

EMERGENCY PROCEDURES

(For the purpose of this instruction the term Principal also means the Senior Person in Charge if the Principal is not available)

1. If possible **NOTIFY** the **PRINCIPAL, HOD OF THE DEPARTMENT** and / or **FIRE FIGHTER** or **SECURITY**, they will decide if it is necessary to **BLOW THE FIRE** emergency alarm.
2. If not possible to notify the persons listed on point 1, below off the emergency alarm.
3. Leave the area and building in a neat, orderly manner to avoid panic and injuries.
4. Follow the evacuation routes specified to the designated and assembly points.
5. Classes / departments should stay in groups together for the roll-call and make sure your name has been checked on the roll call list
DO NOT go anywhere else or return to your work area / classroom for personal belongings.
DO NOT re-enter the premises under any circumstances until the all call alarm has been sounded or until the emergency controller advises you that it is safe to do so.
6. If you are a first aider, take your first aid kit with you when you evacuate from your work area.

My Pledge

*I will be responsible for my safety and
the safety of my colleagues.*

*I will wear the correct PPE when &
where required.*

*I will obey safety rules & follow
procedures.*

I will report all incidents if they happen.

*I will help improve safety in my
workplace and strive to make a
difference.*

My Name:.....Surname:.....Class: Y3

EQUIPMENT: HAND TOOLS

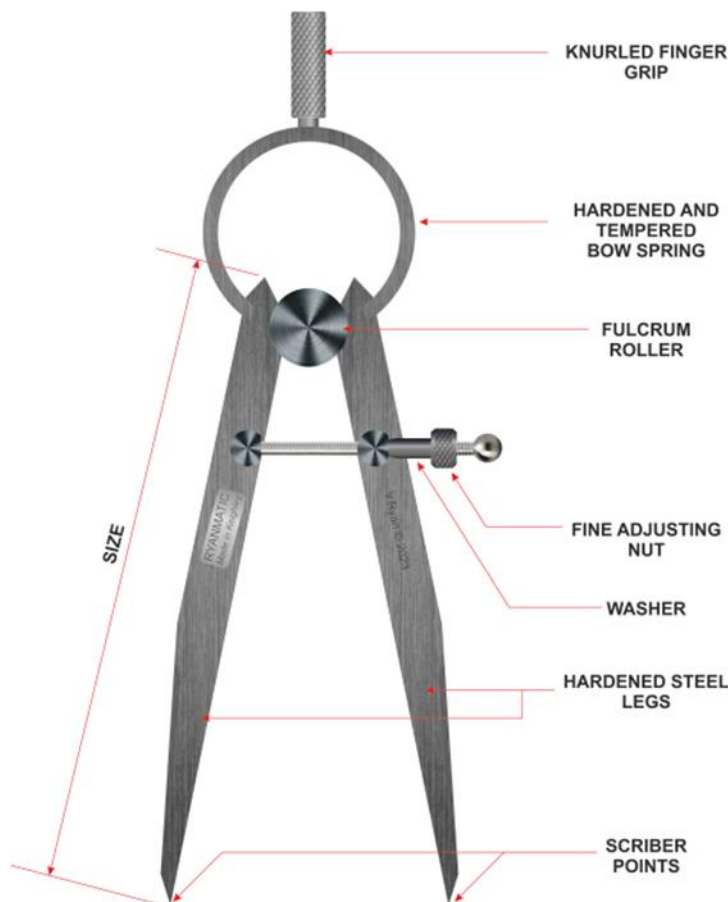
A hand tool is a device for performing work on a material or a physical system using only hands. The hand tools can be manually used employing mechanical force, or electrically powered, using electrical current.

Hand tools:

<https://slideplayer.com/slide/3560282/64/video/Hand+Tools..mp4>

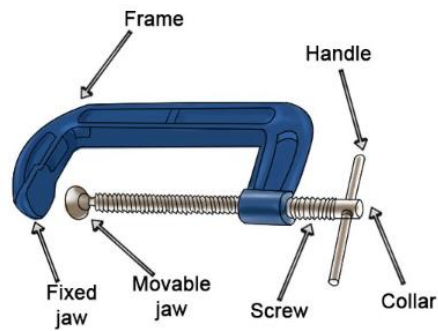
- Dividers

Divider is an instrument for measuring, transferring, or marking off distances, consisting of two straight adjustable legs hinged together and ending in sharp points.



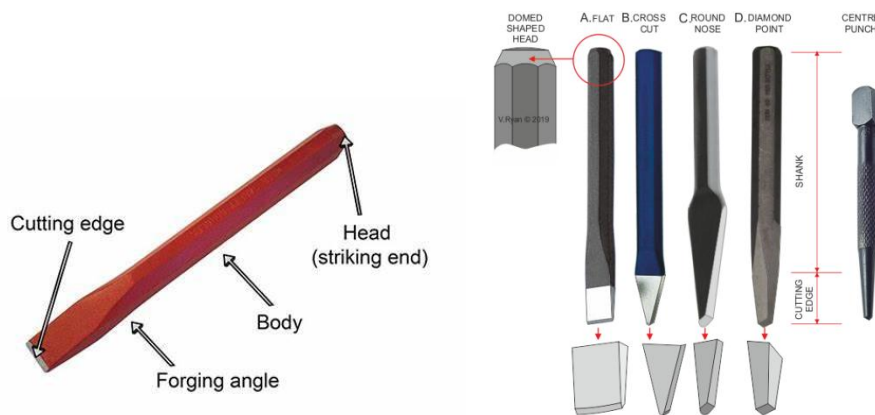
- G clamps

A C-clamp, or G-clamp, is a type of clamp device typically used to hold a wood or metal workpiece, and often used in carpentry and welding.



METAL CHISEL:

Chisel

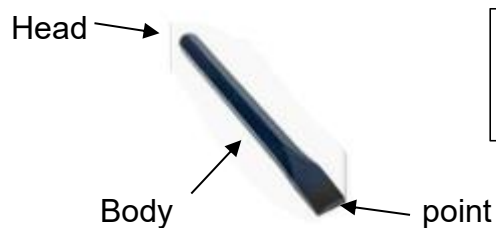


FLAT CHISEL

Use:- To chip and cut flat surfaces, slots and grooves

Material:- High carbon steel or chrome vanadium steel.

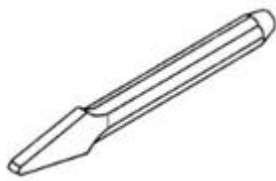
Types:- A. Chisel



B .Hot Chisel

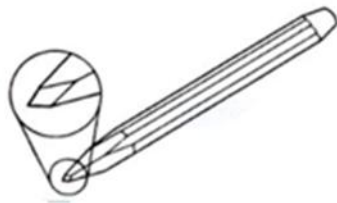
Use:- They are used to remove metal from large flat surface and chip excess metal weld joint and castings

Cross cut Chisel



Use:- These are made for cutting keyways, grooves and slots

Diamond point chisel



Use:- These are used for squaring materials at the corners

Cold chisels work on unheated metal by hammering,

Hot chisel



A hot chisel is used to cut metal that has been heated in a forge or torch to soften the metal. The hot workpiece to be cut is placed over the chisel and struck with a hammer.

ASSESSMENT

Question 1

- a) List the type of chisels that you know.
- b) What are the parts of a chisel?
- c) Which material is used to make a chisel?
- d) What is the use of a chisel?

TYPES OF FILE TOOL AND THEIR USES

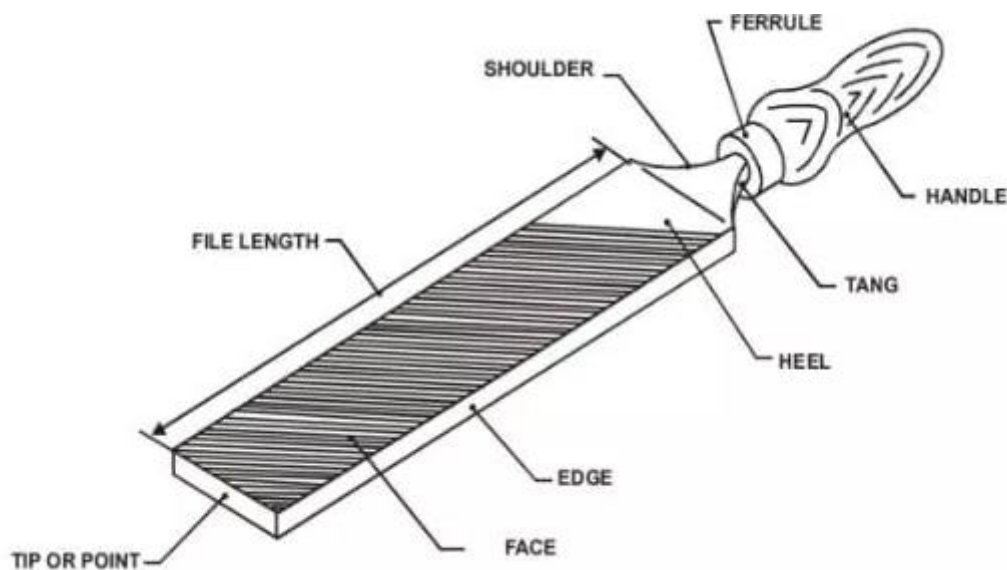
[Types of Metal Files and Their Do's and Dont's \(mechanicalmentor.com\)](https://mechanicalmentor.com)

What is a Metal File?

A **file** is a **tool** used to remove fine amounts of material, burrs, nicks and perform other smoothing operations on a workpiece.

Parts of a File

The main parts of a file include:- point, heel, tooth rows, edge shoulder, and tang.

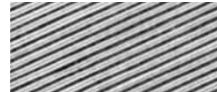


- **Tip/Point:** It is the one end of the file.
- **Face/Side:** It is the wider area that does the filing and contains rows of filing teeth.
- **Edge:** It is the side of the file. In a hand file, one edge has the teeth, and the other does not.
- **Heel:** It is the section of the file face that is free of teeth.
- **Tang:** It is the narrower and thinner part of the file to which is attached handle for holding the file, say by hand.
- **Shoulder:** It is the part of the file where tang is joined with the body of the file. It is curved in shape.
- **Handle:** It is used to hold the file.
- **Ferrule:** It is used to keep the handle from cracking or wearing out. It is a protective ring of metal attached to the handle.

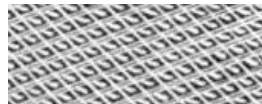
Metal files are made of high carbon steel, alloy steel, tool steel, high speed steel (HSS).

TYPES OF METAL FILES BASED ON TOOTH ART

- **Single-cut file:** If the file has single rows of chisel-cut teeth over its face, it is called a single-cut file. In the single-cut file, there are single rows of coarser teeth that extend across its face and make a 65 to 85-degree angle with the axis of the file.

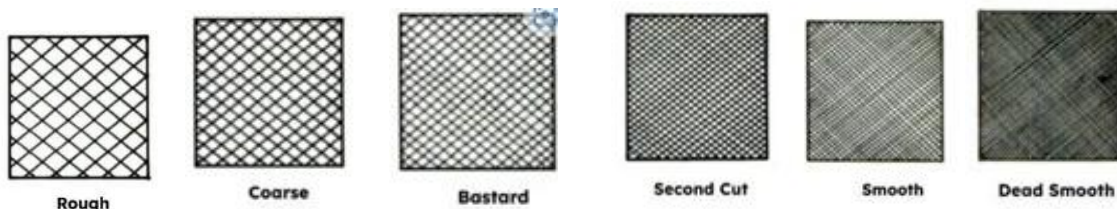


- **Double-cut file:** If the file has double rows of finer chisel-cut teeth in a criss-cross fashion on its face, it is called a double-cut file. For general work, the angle of the first row is from 40 to 45 degrees, while that of the second is from 70° to 80°.



Based on File Grade

Files are broadly divided by the **type of the cut**, which can be rough, coarse, bastard, second-cut, smooth, or dead-smooth

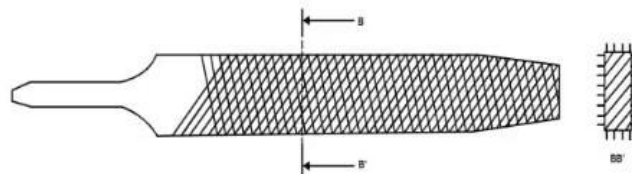


Based on Tooth Geometry

Files are also grouped based on the **geometry of the cut**, such as round, half-round, triangular, and square.

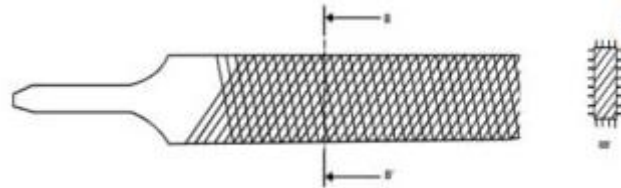
Flat File

It is rectangular in construction. Its two-thirds length is flat, whereas the remaining one-third is tapered towards the point. Its face has double-cut teeth with edges of a single cut-type.



It is the file that is found in every machine shop and is used to file large metal surfaces when the metal in great amounts is to be removed.

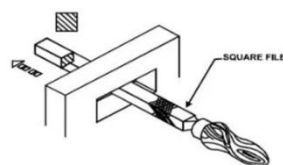
Hand File



It has flat edges that extend from heel to end with its slightly convex surfaces. The second-cut class is for removing feed marks and generating a good surface finish by cleansing. It also has a safe-edge file.

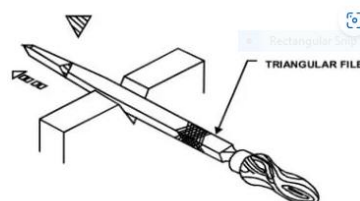
Square File

It is used in flailing square or rectangular holes, as shown below. It is also used in giving surface finish to the narrow bottoms of slots.

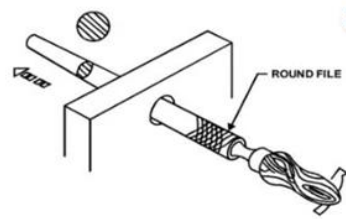


Three Square File

It is also called a triangular file. It has double cuts on all three surfaces with sharp edges. It is used in giving valuable surface finishes, usually less than a right angle.



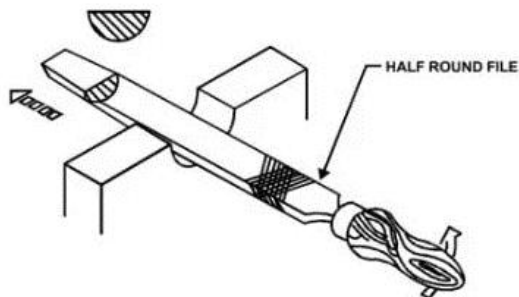
Round File



It finds its use in rounding off the corners. Also, it can be used in enlarging the existing holes.

Half-round File

It is half-round in its construction, as shown below, and is widely available in every workshop. It is made in various sizes, lengths, and design specifications.



Do's in Filing

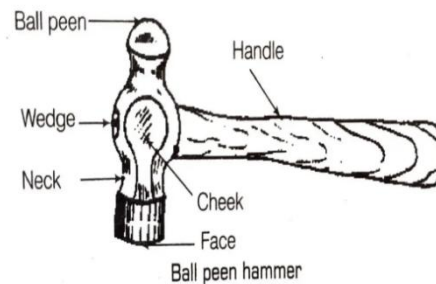
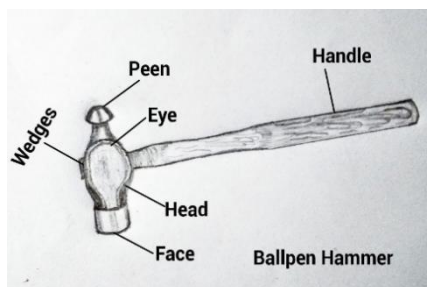
- Over time, filing first is wedged on the teeth of the file. It is known as pinning. It decreases the effective finishing capability of the file. A small piece of soft steel brass or copper should be used to clean the surfaces/edges of the files periodically.
- Keep the right posture during filing: Don't lean too much. Exert arm pressure. Move the file diagonally or straight across the work part.
- The use of chalk can also clean the teeth from clogging. It should be a practice after completing a particular job.

Don'ts in Filing

- Don't use a file without a handle.
- Don't use a worn-out or dirty file.
- Don't use a new file on the cast iron scale as well as on narrow edges.
- Don't use a file for which it is not designed.
- Don't push the file too fast.
- Never hammer on or use it as a pry bar. A file is very brittle and will break easily.

USE A FILE CARD TO CLEAN A FILE.

- **Ball Peen/Pein Hammer Also Known As A Machinist Or Engineer's Hammer**



What is A Ball Peen Hammer?

is a type of **peening hammer** used in **metalworking**.

The ball-peen hammer is useful for many tasks, such as striking **punches** and **chisels** (usually performed with the flat face of the hammer).

The peening face is useful for rounding off edges of metal pins and fasteners, such as **rivets**.

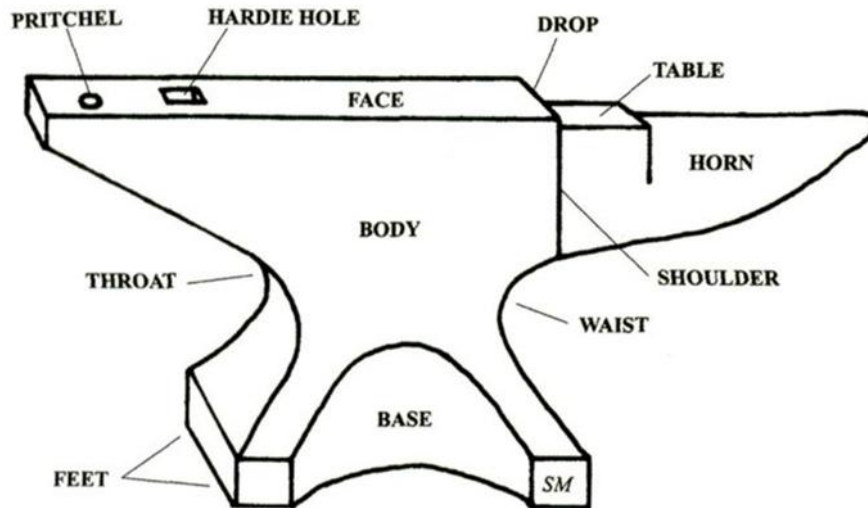
[All About Ball Peen Hammer | Hand Tools Course | SkillCat \(youtube.com\)](#)

What is A Ball Peen Hammer Used For?

1. **Shaping Metal**
2. **Armor, Sword, and Knife Making**
3. **Peening Rivets**

ANVIL

They are five main parts on anvil:



The Face

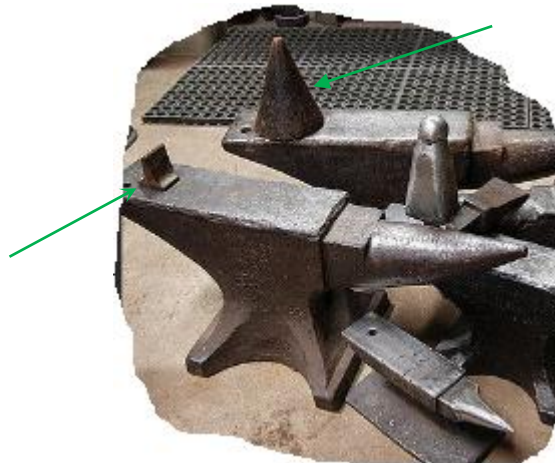
The face is hardened steel area where most work is done.

The Step/table/chipping block

is between the horn and the face, used for cutting to avoid damaging the face.

The Hardy Hole

Square hole on the tail for mounting tools such as chisels and swages



The Pritchel Hole

Is just a round hole on the tail; used for punching work(making holes



SPANNERS

Ensure that the spanner is the correct one for the particular bolt or nut. The tool must fit over the bolt head or nut with minimum of free play.

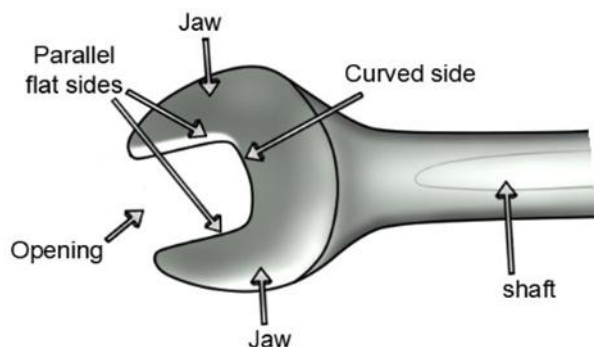
NB: As far as possible, “**PULL**” the spanner don’t “**PUSH**” it.

When you have to push the spanner, do so with an open hand and push in the direction in which force is applied.

THERE ARE FOUR TYPES OF SPANNERS

- Open-ended spanner
- Ring spanner
- Combination spanner
- Socket spanner

Open-ended spanner



Open – ended spanner is for general work where it is impossible to use a ring spanner (because of a confined space). Be careful when tightening a bolt as the open may stretch and cause the spanner to slip.

Ring spanner



Use a ring spanner to loosen or tighten bolts and nuts. There are two advantages of using this kind of spanner:-

- a) More teeth come into contact with the head of the bolt or nut
- b) The spanner is not likely to slip

Combination spanners



A combination spanner has a ring spanner at one end and open – ended spanner on the other end.

SOCKETS

Sockets are used to tighten or loosen bolts and nuts to save time. You can also use them in situation when you cannot reach nuts or bolts with ordinary spanners.



A socket is used with various accessories such as:

- a speed wrench or brace
- flexible joints
- universal joints
- a ratchet
- various extensions
- power bars

- 1) Sketch three different types of spanners.
- 2) When should you use an open – ended spanner?
- 3) Which accessories can be used with ratchets, and specify where each of them can be used?

JIGS

What Is a Jig?



A jig is a large brace that keeps a welding project stable in the face of pressure, heat, motion, and force. A quality jig will streamline welding work by keeping parts together in a vice grip.

Types of Jigs

Following are the seven different types of jigs.

4. Diameter jig
5. Leaf jig
6. Ring jig
7. Box jig

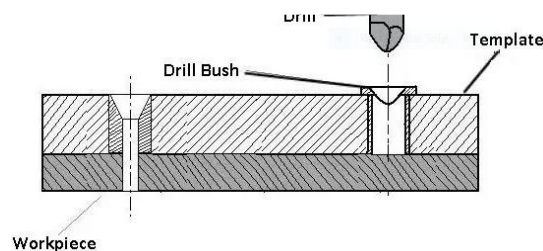


Plate Jig

1. A jig holds, locates, and as well as guides the tool.
2. The jigs are made lighter for quicker handling, and clamping with the table is often unnecessary.
3. The jigs are used for holding work and guiding the tool, particularly in [drilling, reaming, or tapping operations](#).

Advantages of Jigs are:

1. The handling time is also greatly reduced due to the quick setting and locating of the work.
2. The speed, feed and depth of cut for machining can be increased due to high clamping rigidity of jigs and fixtures.
3. These devices reduce the operator's labour and consequent fatigue handling operations, are minimized and simplified.

Disadvantages of Jigs

The following are the disadvantages of jigs and fixtures:

1. Over time, jigs may become worn out.
2. The second disadvantage is the significant cost and time required for initial setup.

FINISHING GRINDER



A finish grinding machine is a specialized industrial equipment that is used to grind and polish surfaces with great precision and smoothness.

1. Metal

Grinding machines are common in metalworking applications. They can cut, shape, and finish various types of metals, including: carbon steel, stainless steel, aluminum, brass and iron.

[Surface grinder - finished \(youtube.com\)](https://www.youtube.com/watch?v=...)

HEAT GUN



Heat guns are devices that emit a stream of hot air, usually at temperatures between 100°C and 550°C. They are used in various industries and projects for different purposes. Some common uses of heat guns include:

- Loosening old bolts and screws
- Bending plastic and metal
- Removing adhesives
- Embossing
- Waxing

- Shrink wrapping large objects
- Drying out damp wood or other materials prone to molds.
- Sterilizing tools for proper sanitation
- Welding plastic and vinyl wrap

- Paint stripping

Keep in mind the following safety tips:

- Wear the proper safety gear
- Ensure good ventilation
- Remove combustible/flammable materials
- Check temperature settings are correct
- Do not point the gun at one spot for too long
- Switch the tool OFF before putting it down on any surface
- Never touch the hot metal nozzle with clothing or skin
- Never direct the airflow towards yourself or another person
- Do not use with lead-based paint to avoid toxic fumes
- Never block the inlet grill or obstruct the airflow of the unit while in operation

Type of Heat Gun	Characteristic	Usage
Electric Heat Guns	Powered by mains electricity; Temperature range: 100°C to 550°C	Household tasks: stripping paint, thawing pipes, crafting
Gas Heat Guns	Powered by butane/propane gas canisters; Temperature up to 600°C	Outdoor use: roofing tasks, field repairs, camping

WHAT IS A SHEET METAL BOX AND PAN FOLDER?

BOX AND PAN BRAKE



It is called a box and pan brake because the machine is designed to form, bend and fold sheet metal into boxes and pans, as well as other shapes.

The machine fingers or jaws are interchangeable and can be assembled in any sequence, combination or place in the clamping beam.

Care must be taken:

- 1) Before use check that the machine is mounted securely.
- 2) **Not to go above the indicated limit (metal thickness) capacity of the machine to bend sheets.**
- 3) Only bend sheet metal, no rods or angle iron on the machine.
- 4) To use only arm and legs power, not the back or extensions to the folding bar levers.

UNIVERSAL SCROLL BENDER





Std Scroller and Scroller Base



Actual Scroll Patterns

Left to Right: Oversize, Large, Medium, Standard



Scroller-Instructions.pdf

	Standard scroller	Medium scroller	Large scroller	Oversize scroller
Scroll diameter	155mm	155mm	205mm	25mm
Flat bar	40.5mm	30.8mm	30.10mm	30.10mm
Square bar	8mm	12mm	16mm	16mm
Round bar	8mm	12mm	16mm	16mm
Square tube	n/a	12mm	16mm	20mm
Round tube	n/a	12mm	16mm	20mm

METAL TWISTERS



TWISTER-VIDEO-CLIP-2018.mp4

<https://youtu.be/9eFT4sXIZKA>

What is a metal twister?

A compact tool for twisting flat and square metal for decorative and structural projects.

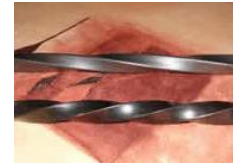


Twist-Instructions.pdf

https://youtu.be/FQeoD8H2_H0

<https://youtu.be/0gb7iaGITIE>

<https://youtu.be/rVshiEucpEE>



BENEFITS

Twists are perfectly straight.

Requires no heat.

Fully adjustable-can twist an entire 6m length

- Metal shearing.



This machine operates like a pair of scissors, the operator must guide the metal plate carefully to make sure that the cut follows the marked.

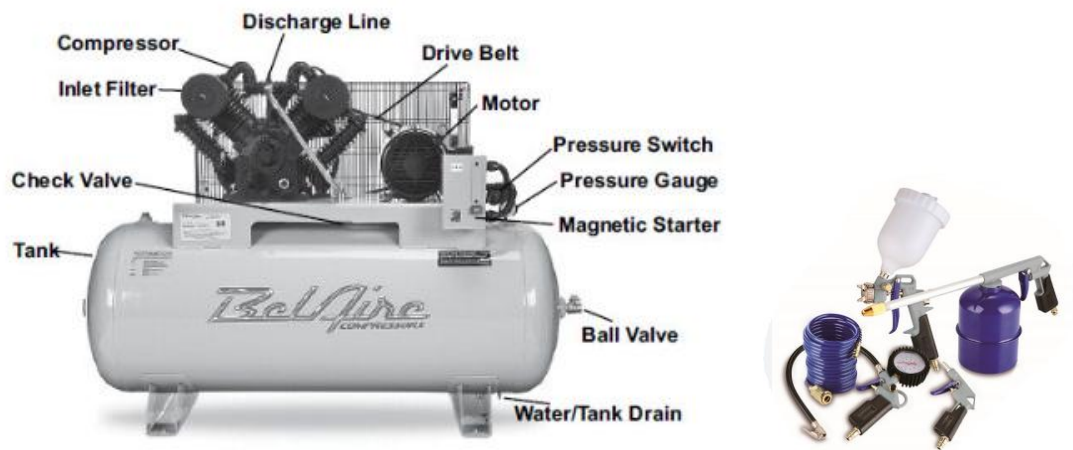
Safety precaution when working on the shearing machine

- Care must be taken to keep hands and fingers away from the blades.
- Keep onlookers and passers-by out of reach of the long handle as the handle has a tendency to drop unexpectedly once it cuts through the material.

Power hand tools

Compressor

A compressor is a device used to compress air.



What is the compressor used for?

A compressor is a device used to compress air or other gases to pressures higher than atmosphere pressure. It compresses the air taken from the atmosphere and increases its pressure

Air compressor maintenance

Drain Tanks Moisture

- If you live in a humid environment, the air compressor tank will collect moisture faster.
- Make sure water is drained regularly but keep in mind to release the air first before draining the water.

Check and Change Air Filters If Needed

- A heavy build-up of dust and dirt in your air compressor's air filter may cause it to take in some of these residues. When this happens, It lowers the output capacity of the power tool. Keep the Intake Vents Cleaned
- Regularly check the intake vents of your compressor to make sure that they are clean.
- Check and Change Compressor Oil
- Compressor oil, acts as a lubricant and a cooling agent that prevents your air compressor to overheat easily and prevent its parts from wear and tear Regularly

Check the Safety Shutdown System on Screw and Valve Type Air

Compressors

- A safety shutdown system is built within air compressors so it automatically shuts off when the compressor overheats or if the pressure is too low. Test if the system works so you can ensure a longer lasting compressor.

Types of Air Compressors

The following are the types of air compressors:

1. Reciprocating air compressor
2. Rotary air compressor
3. Vane-type air compressor
4. Centrifugal air compressor
5. Axial air compressor

Pneumatic Hand Tools

- Hand drill
- Finishing grinder
- Heat gun
- SDS rotary drill.

Pneumatic Tools or Air Tools are those power tools that use compressed air as the main source of energy.

Advantages and disadvantages

The benefits of using compressed air to power tools are:

- Inexpensive
- Safe to use
- Easy to operate
- Portable
- Low theft rates

The primary disadvantage of pneumatic tools is the need for an [air compressor](#), which can be expensive. Pneumatic tools also need to be properly maintained and oiled regularly. Failing to maintain tools can lead to **deterioration**, due to a build up residual oil and water.

Air Impact Wrench



Air Wrenches are very efficient and can remove/fix nuts and bolts very quickly. They have a very high torque and can remove even stuck, stubborn and delicate nuts and bolts very easily.

Pneumatic Drill 	PAINT SPRAYER 
Another important application of compressed air is spray painting.	

Pneumatic Reciprocating Saw

We can cut through tiles, sheet metal, pipes (steel, copper), fiberglass etc.

Conclusion

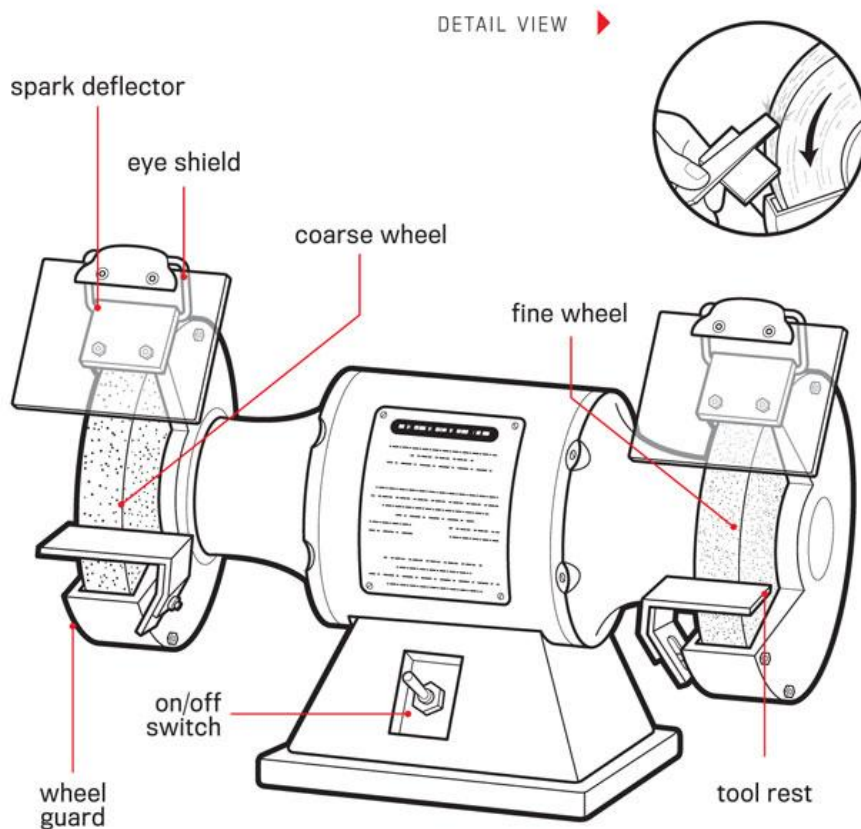
Air Tools make the job easy and fast, whether it is drilling holes, tighten/loosen nuts/bolts, demolishing roads/building, inflating tires, painting etc. They are also known as Pneumatic Tools as they use compressed air as the source of energy (in contrast to electricity of fuel). We saw the basics of air tools, how they work and also different types of air tools along with their uses

FLOOR MOUNTED EQUIPMENT:

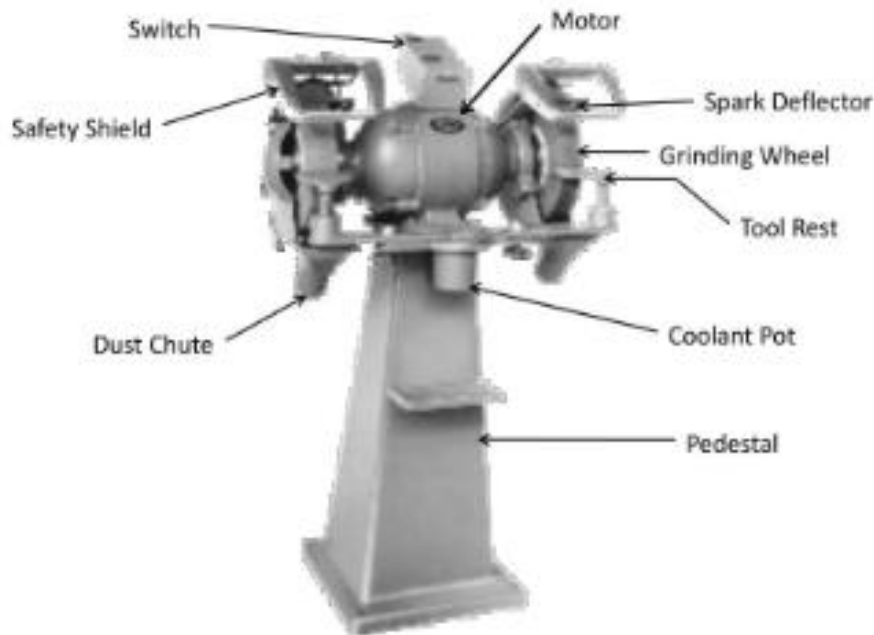
BENCH GRINDER

These types of grinders are commonly used to hand grind various cutting tools and perform other rough grinding.

A bench grinder is the perfect power tool to shape up steel objects by sharpening. A **bench grinder** is a **bench top** type of **grinding machine** used to drive **abrasive wheels**.



A **pedestal grinder** is a similar or larger version of grinder that is mounted on a [pedestal](#), which may be bolted to the floor or may sit on [rubber feet](#).



Bench Grinder Safety: What to Wear for Protection?

- You must wear safety glasses all the time
- Tie up your hair if it's long and loose
- Wearing a hearing protector is a must
- Wear protective clothing at all times

Pre-Operational Safety

- Check the machine's gap between the wheel and the rest is no less than 1.5 mm and the gap must not exceed 3mm.
- Check the wheels are stable and running well

During Grinding Operation

- Before turning the machine on, stand beside the wheels. Let the machine idle for few seconds.
- Only one person can operate the bench grinder at a time
- Use clamping pliers when you can't hold the grinding parts by hand

Post-Operational Safety

- Turn the machine off if you are done with your work
- Wait until it stops completely
- Clean the work area and look for coolant spills to remove
- Never try to use anything to stop the grinder

Hazards and Injuries

- Parts of the object you are grinding may get sucked into the machine
- A damaged wheel may explode anytime, so replace it

What is a Pedestal Drill?

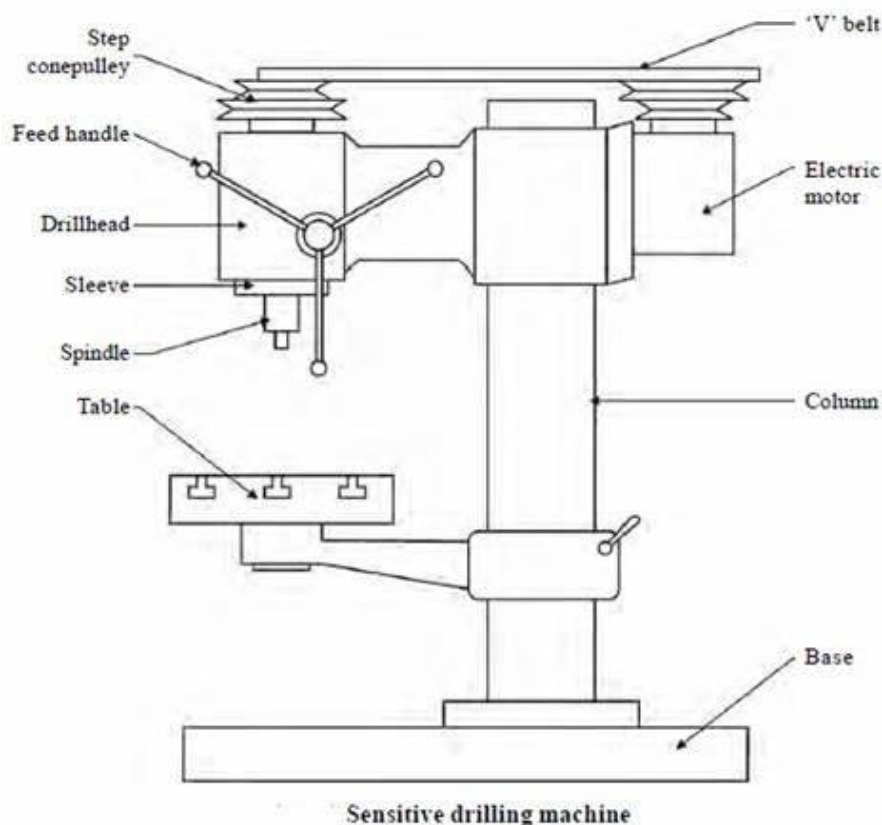
A pedestal drill is a power tool that can be mounted to a stand or bolted to a workbench or the floor.

It is often referred to as a drill press as well as a pillar drill or a bench drill.

This drill is used to drill holes **up to 12.5 mm in diameter**.

Safety precautions when using drill press

- Do not leave the chuck key when you are not at the machine.
- Never leave the machine running if it is unattended.
- Clamp the work piece securely to the table and do not hold it by hand.
- Never try to stop the work piece by hand if it slips from the clamp.
- Do not force a drill bit into the work piece- this may cause drill bit to break.



What are the Parts of a Pedestal Drill?

The Base

The base is the piece that supports the drill and stabilizes the column.

The Column

The column (sometimes called the pillar) is positioned vertically and supports the table, fixing it securely to the base.

The Table

The table supports the material (plastic, metal, wood, etc.) and can be adjusted to fit accordingly

The Spindle

The spindle is the rotating piece that holds the drill bit in place. Its speed can be adjusted to control both the diameter and depth of the hole being cut.

The Chuck

The chuck is a type of 3-jawed clamp which is mounted on the spindle and fits securely around the shank of the drill bit. A chuck comes in multiple sizes.

The Drill Head

The drill head supports the spindle, allowing it to move either up or down.

300MM DIAMETER DISC SANDER

What's Sanding?



Sanding is smoothing or polishing a surface with sand or sandpaper. Alternatively, you may use steel wool instead of sandpaper for fine-sanding and polishing projects

To use a 300mm diameter disc sander for metals, follow these general steps:

How to Sand Metal for Painting: A Step by Step Guide

SAFETY FIRST

- Wear a dust mask
- Wear safety goggle
- Remove clutter and obstructions.

Requirements

- Wire brush attachment
- Sander
- Sandpapers in different grits
- Dry cloth or rags

How to Sand a Metal Surface: Step-by-Step

1. **Scrub the entire surface with a wire brush to remove old paint:**
This helps to loosen paint, and a thorough scrub with a wire brush removes loose rust.
2. **Continue until you strip the surface to bare metal:**
This is critical whether or not you intend to repaint the surface.
3. **Begin with coarse grit sandpaper:** Coarse grit sandpapers help to remove heavy dirt buildup. For instance, begin with 80-grit sandpaper for rusted metal. Alternatively, start with 120-grit sandpaper.

4. **Work your way up:** The best way to sand metal is to gradually up the grit levels without skipping too many stages. For instance, don't jump from 80-grit sandpaper to 220-grit sandpaper, as it creates a poor finish. Instead, move from 80 to 120, and then 180 before getting to 220.
5. **Polish and buff the surface if needed:** You can stop at 220 grit sandpaper if you intend to repaint. It gives a beautifully painted surface. Otherwise, polish and buff the surface with 400-grit sandpaper and then 600-grit sandpaper for a shiny finish.

Summary

Sanding is a critical metal restoration and finishing process. It removes rust, eliminates rough edges, and evens surface imperfections. Therefore, it pays to understand the different metal sanding tools, sandpapers, and techniques.



- Guillotine.

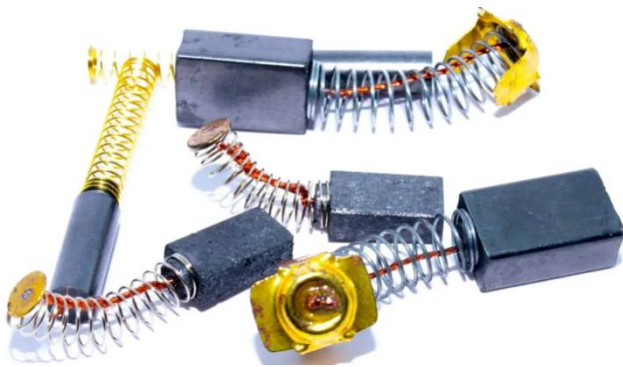
Guillotine sheet metal cutting machines cut metal sheets into specific shapes, sizes, and dimensions. You can use these machines to cut metal sheets with absolute accuracy.

Educator to demonstrate the correct way in which power tools need to be stored and also the way electrical cabling needs to be properly rolled up.

- **REPAIR AND REPLACE A FAULTY 3 CORE CABLE.**
https://www.youtube.com/watch?v=axUU_Q2ef_4
<https://www.youtube.com/watch?v=qHoyF4yKhjU>

Replacing brushes and bearings inside power hand tools and motors.

How To Replace Carbon Brushes In Your Power Tool



1. First off, make sure that the power tool is unplugged.
2. Remove the motor from the power tool. In most power tools, you simply need to remove the cover of the motor. You'll need to undo bolts or screws which might be holding it in place.
3. Find the carbon brush of the motor. Most carbon brushes are small, rectangular parts with a connected spring and wire and a metal part at the other end.
4. Once you find it, remove its electrical wires. Pull it from its housing and pay attention to what direction it's in.
5. Simply insert the new carbon brush and do the steps in reverse.

As always, when handling complex tools, be extra careful when replacing certain parts.

► [How To - Replace Worn Power Drill Brushes](#) ◀ ([youtube.com](#))

Wire A Plug In South Africa

- a) To wire a plug in South Africa, start by unscrewing the cover of the plug to access the terminals.
- b) Place the cord into the plug and ensure that the line and neutral wires are connected to the correct terminals.

- c) The right terminal should be marked with a blue line or an N, while the left terminal should be marked with a brown line or a L. Secure the wires in place by tightening the screws on each terminal.
- d) You can then replace the plug cover and test the plug to make sure the connection is secure.
- e) Finally, plug the plug into a wall socket and make sure the appliance works correctly.

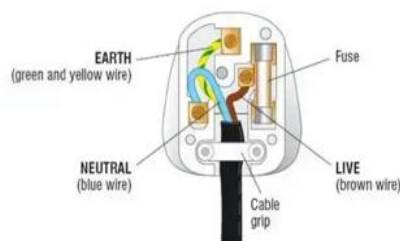
<https://aswica.co.za/how-to-wire-a-plug-in-south-africa-step-by-step/>



<https://aswica.co.za/how-to-wire-a-plug-in-south-africa-step-by-step/>

Step-by-step guide on wiring the plug

Step 1: Gather Materials



Before you begin your wiring project, make sure you have all the necessary materials. You will need a plug and the corresponding wiring, a screwdriver, and a pair of wire strippers.

Step 2: Strip the Wires

Using your wire strippers, carefully strip the ends of



the wiring. Make sure to remove approximately 10 millimeters of the insulation. Expose the copper core of the wires and, if necessary, twist them together.

Step 3: Connect the Wiring



Now it's time to connect the wires to the plug. Connect the live (brown) wire to the live terminal and the neutral (blue) wire to the neutral terminal. Make sure the wires are correctly connected to their corresponding terminals.

Step 4: Secure the Wiring

Use your screwdriver to tighten the screws on the terminals. Make sure to tighten the screws firmly enough for the wires to stay in place, but be careful not to over tighten them.

Step 5: Test the Connection

To test your wiring, plug the appliance into a power outlet. If the appliance turns on, your connection is successful. If not, double check your wiring to make sure it's

MATERIAL SELECTION

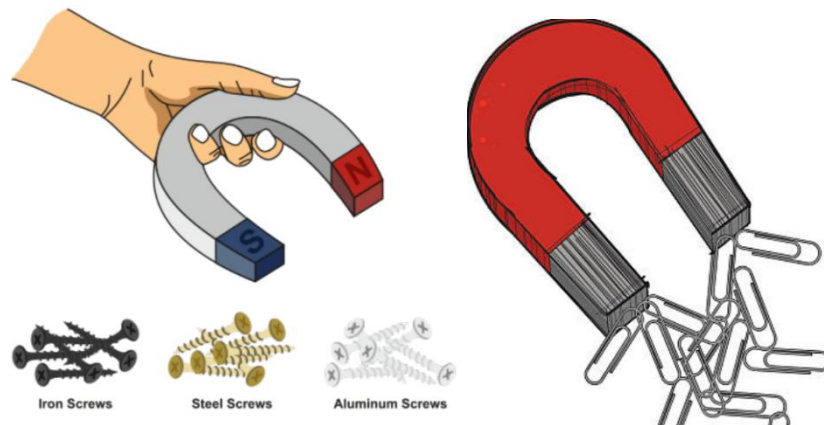
MAGNETIC TEST

use a magnet

SPARK TEST

use a grinder

Magnetic Testing!



This is the most commonly used method for testing metal.

ferrous metal are alloys with an iron base and they are usually magnetic.

Nonferrous metal is not magnetic.

Using a magnet is not one hundred percent accurate due to the fact that there are some stainless steel that is not magnetic.

What Is Spark Testing?

Spark testing is a method of determining the general classification of ferrous materials. It normally entails taking a piece of metal, usually scrap, and applying it to a grinding wheel in order to observe the sparks emitted.

To perform a spark test on mild steel, you need to:

1. Put on personal protective equipment.
2. Use a bench grinder or portable grinder with a speed of 4500 feet-per-minute.
3. Hold the steel against the grinder so the sparks fly across your field of vision.
4. Check the length and form of the sparks for indications of mild steel.
5. Look for some variety in spark streak length, which indicates mild steel.

Mild steel spark test

- Both low carbon and high carbon steel produce white spark streams.



High carbon steel

<https://youtu.be/7PWCh6fdXdw>

<https://youtu.be/D094eBa4S7c>

<https://youtu.be/GnSBSKTC834>

Make sure you follow all the safety guidelines associated with using a grinder. These tools are very handy but they are also very dangerous.

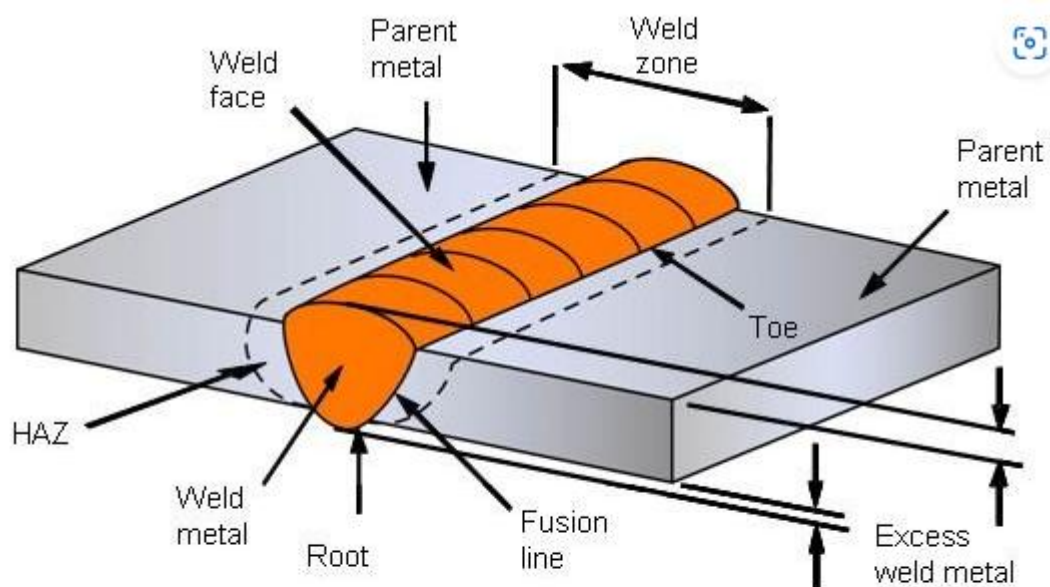
Metals	Colour unfinished, Unbroken surface	Colour and structure of newly fractured surface	Colour filed surface
Gray cast iron	Dull gray	Dark gray crystalline	Light silvery gray
Wrought iron	Light gray	Bright gray	Light silvery gray
High carbon steel	Dark gray	Light gray	Bright silvery gray
Stainless steel	Dark gray	Medium gray	Bright silvery gray
copper	Reddish brown to green	Bright red	Bright copper colour
Brass and bronze	Reddish yellow, yellow,-green or brown	Red to yellow	Reddish yellow to reddish white
aluminum	Light gray	White; finely crystalline	white

- Materials to be identified is limited to mild steel, aluminium, stainless steel and carbon steels.



Explain the following joints:

- Butt joint.



Butt weld diagram

A butt weld is one of the simplest and versatile types of weld joint designs. The joint is formed simply by placing two pieces of metal end-to-end and then welding along the join. Importantly, in a butt joint, the surfaces of the workpieces being joined are on the same plane and weld metal remains within the planes of the surfaces. Thus, workpieces are nearly parallel and do not overlap, unlike lap joints, for example.

Advantages

Butt welds provide several advantages, including:

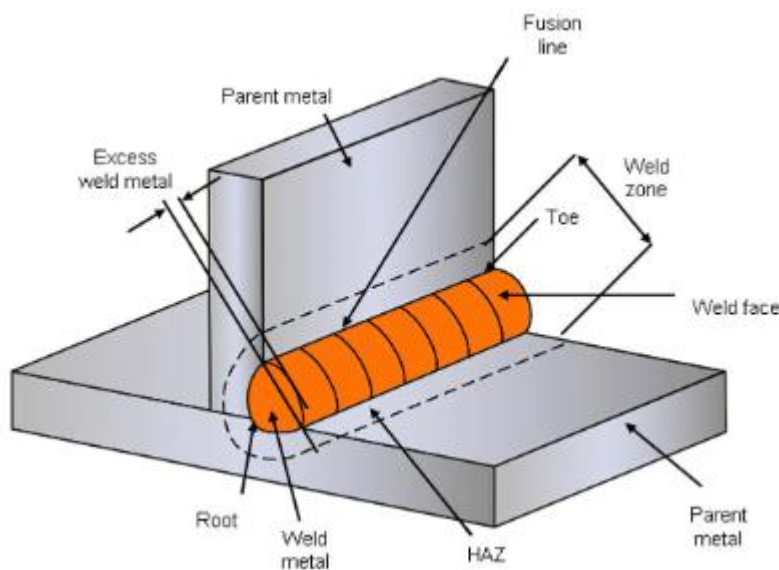
- High strength with complete fusion
- Easy to machine
- Distortion control
- Easy to inspect

Disadvantages

There are some disadvantages such as:

- Welding geometry can limit applications
- Fixturing or backing may be required
- Sensitive to faying surface conditions

- Fillet joint (T joint)



Lap joint

What is Welding Joint?

Methods of Joint

A welded joint is created by joining two materials together that are similar or dissimilar while applying pressure and filler metal, if necessary. There are two categories of welding, depending on how the joint is prepared:

1. **Fusion Welding:** In this method of welding, the welding joint is created by melting the parent metal.
2. **Non-fusion Welding:** This method of welding creates the joint without melting the parent metal.

Types of Welding Joints

A proper weld joint must be made according to the requirement, assembly, design, and operating conditions. It, therefore, requires various types of welded joints depending on the above parameter.

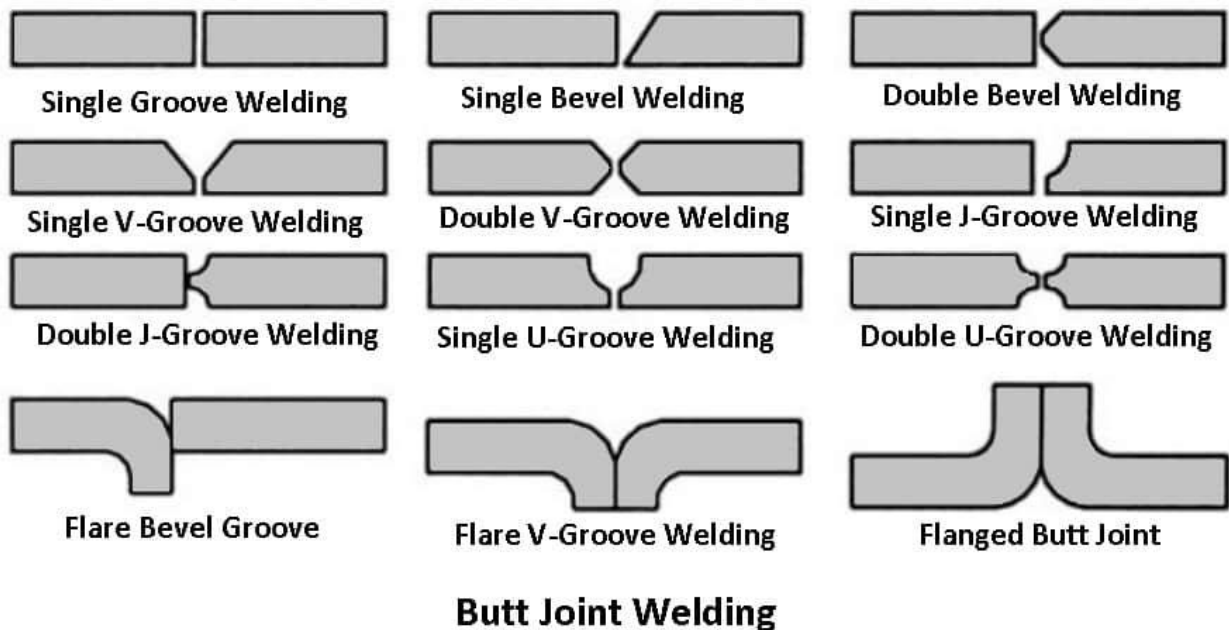
Following are the 5 main **types of welding joints**:

1. Butt joint welding
2. Lap joint welding
3. Edge joint welding
4. Tee joint welding
5. Corner joint welding

#1 Butt Joint Welding

These types of welding joints form when the two metal pieces are placed end to end together in the same plane surface, it is known as butt joint welding.

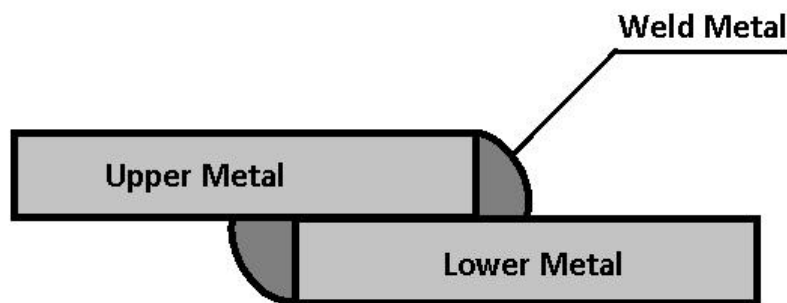
Following are some typical examples of butt weld joints that are included in welding.



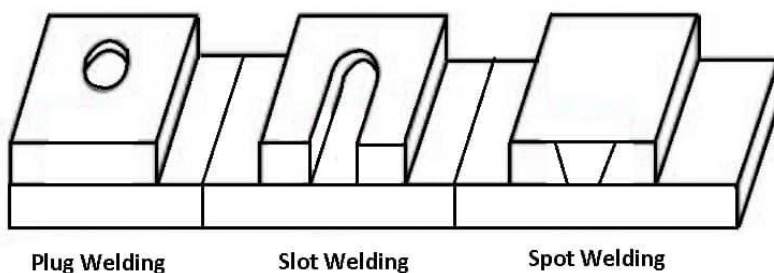
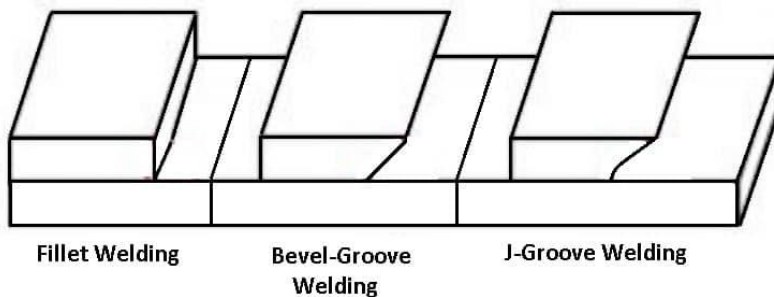
1. Square welding

2. Single bevel welding
3. Double bevel welding
4. V-groove welding
5. Double V-groove welding
6. U-groove welding
7. Double U-groove welding
8. J-welding welding
9. Double J-groove welding

#2 Lap Joint Welding



It is basically a modification of butt welding, in which the two metal plates are placed in an overlapping position that is on top of each other, it is known as lap joint welding.



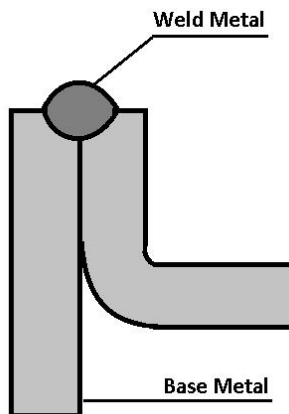
Lap Joint Welding

1. Fillet welding
2. Spot welding
3. Plug welding
4. Slot welding

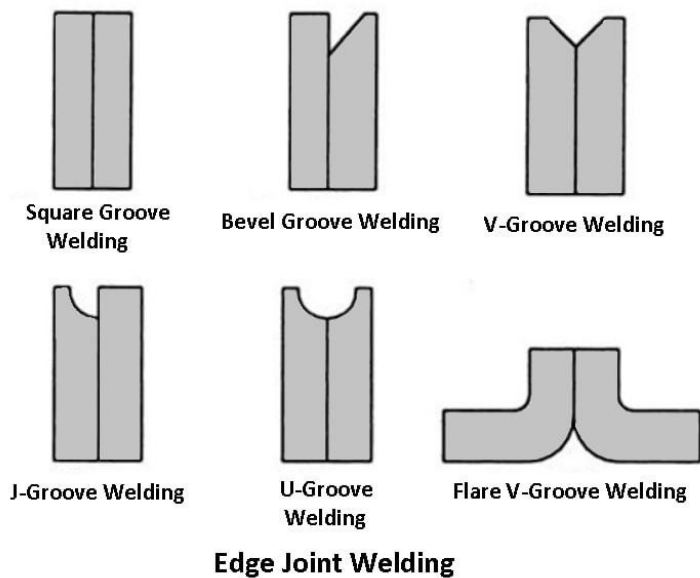
5. Bevel groove welding
6. Flare bevel groove welding
7. J-groove welding

You might like: [Understand The Different Types of Welding Machines & Their Uses](#)

#3 Edge Joint Welding



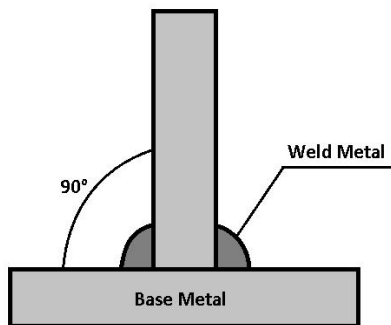
This is quite similar to the lap joint.



1. V-groove welding
2. U-groove welding
3. J-groove welding
4. Bevel groove welding
5. Square groove welding
6. Corner flange welding
7. Edge flange welding

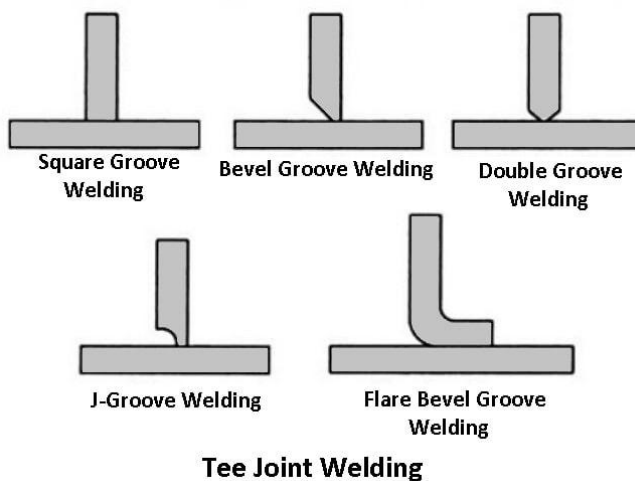
You might like: [26 Essential Welding Tools and Equipments \[How To Use Guide\]](#)

#4 Tee Joint Welding



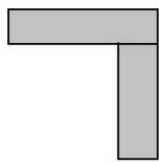
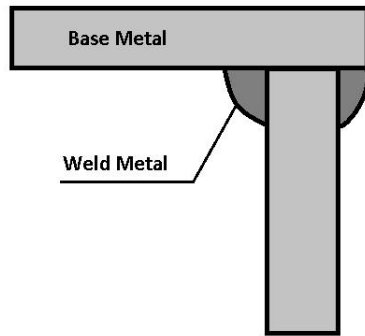
It is formed, when the two metal plates are intersected to an angle of 90° with one plate is lie on the center of the other plate like a “T” shape. It is known as tee joint welding.

Following are examples of tee joint welding.

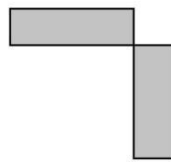


1. Plug welding
2. Slot welding
3. Bevel groove welding
4. Flare bevel groove welding
5. Fillet welding
6. J-groove welding
7. Melt through welding

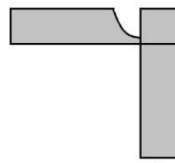
#5 Corner Joint Welding



Spot Welding



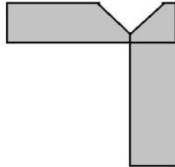
Fillet Welding



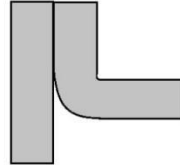
J-Groove Welding



Edge Welding



V-Groove Welding



Corner Flange Welding

Corner Joint Welding

1. Fillet welding
2. Spot welding
3. Edge welding
4. J-groove welding